Math 104 – Homework 1 Answers

Homework 1

Deadline: 11:59 pm September 6, 2026

Problem (1)**.** *Guess a formula for*

$$1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2$$

(the sum of the squares of the first n odd numbers) by evaluating the sum for $n = 1, 2, 3, 4$. Then prove your formula by mathematical induction.

Problem (2)**.** *Determine* all *positive integers n for which*

$$3^n > n^3.$$

Prove your claim by mathematical induction. Be careful: unlike the original exercise, the inequality is not *monotonic in the small cases, so check $n = 1, 2, 3$ directly before deciding where an inductive argument can safely begin.*

Problem (3)**.** *Prove that $\sqrt{2} + \sqrt{3}$ is irrational.*

Hint: Suppose $\sqrt{2} + \sqrt{3} = r \in \mathbb{Q}$, isolate one of the radicals, and square.

Problem (4)**.** *For $a, b \in \mathbb{R}$, prove that*

$$|a+b| = |a| + |b| \qquad \text{if and only if} \qquad ab \geq 0.$$

(That is, equality holds in the triangle inequality exactly when a and b have the same sign, or at least one of them is 0.)

Problem (5)**.** *Prove by induction that for all integers $n \geq 2$ and all real numbers $a_1, a_2, \ldots, a_n$,*

$$|a_1 - a_2 - a_3 - \cdots - a_n| \geq |a_1| - |a_2| - |a_3| - \cdots - |a_n|.$$

Hint: First establish the case $n = 2$, i.e., $|a - b| \geq |a| - |b|$ for all $a, b \in \mathbb{R}$, as a lemma from the ordinary triangle inequality; then use it in the inductive step.

Solutions

Problem 1

Evaluating the first few, we get $1, 10, 35, 84, \cdots$

Claim:

$$\sum_{i=1}^{n}(2i-1)^2 = \frac{1}{3}(n)\left(4n^2-1\right)$$

We proceed via induction, denoting $P(n)$ as the truth value of this equality at n.

Our base case, $n = 1$, holds, as $\frac{1}{3}(1)(4-1) = 1$

Inductive Hypothesis: We claim that this identity holds for $1, 2, 3, \cdots, k$ for an arbitrary $k \in \mathbb{Z}^+, k \geq 1$

Written out, this identity is

$$1^2 + 3^2 + \cdots + (2k-1)^2 = \frac{1}{3}k(4k^2-1)$$

Adding the next term, $(2k+1)^2$ to both sides yields on the LHS:

$$1^2 + 3^2 + \cdots (2k+1)^2$$

and on the RHS:

$$\frac{1}{3}k(4k^2-1) + (2k+1)^2$$

The RHS can be expanded, factoring out the $\frac{1}{3}$ and then expanding the second term, yielding

$$\begin{aligned}
&\frac{1}{3}\left(4k^3 - k + 12k^2 + 12k + 3\right) \\
&= \frac{1}{3}\left(4k^3 + 12k^2 + 11k + 3\right) \\
&= \frac{1}{3}\left(k+1\right)\left(4k^2 + 8k + 3\right) \\
&= \frac{1}{3}\left(k+1\right)\left(4(k+1)^2 - 1\right)
\end{aligned}$$

Solution notes

Problem 2

Solution notes

We claim this inequality holds for $n = 1, 2$ and positive integers greater than 3.

Notice $3^1 > 1^3$ and $3^2 > 2^3$, and that $3^3 = 3^3$, showing that this holds for $1, 2$ but not 3.

We can now examine this inequality for positive integers greater than 3, and proceed via induction.

In this scenario, denote $P(n)$ as this inequality's truth value for n, our base case would be $P(4) \to 3^4 > 4^3$, which is true.

Base Case Holds

Inductive Hypothesis: We claim that $P(i)$ holds for $i = 4, 5, \cdots k$

Multiplying $P(k)$ which is $3^k > k^3$ by 3 yields

$$3^{k+1} > 3k^3$$

Claim: $3k^3 > (k+1)^3$ for $k \in \mathbb{Z}^+ \; k \geq 4$

The simplest way to show this is to see that for $k \geq 4$ that $k + 1 \leq \frac{5}{4}k$ and trivially we have that

$$\left(\frac{5}{4}k\right)^3 \leq 3k^3$$

because $\frac{125}{64} \approx 1.95$ is less than 3. Thus, we have

$$3k^3 \geq \left(\frac{5}{4}k\right)^3 \geq (k+1)^3$$

And remembering our inductive hypothesis, we have

$$3^{k+1} > 3k^3 \geq \left(\frac{5}{4}k\right)^3 \geq (k+1)^3$$

and we are done as we've shown $P(k) \implies P(k+1)$

Problem 3

Solution notes

We proceed via contradiction.

Claim: $\sqrt{2}+\sqrt{3}$ is rational, and can be represented as a fraction in lowest terms as $\frac{a}{b}$ for

$$a, b \in \mathbb{Z}+, \gcd(a, b) = 1$$

Let's square both sides, to get

$$\frac{a^2}{b^2} = \left(\sqrt{2}+\sqrt{3}\right)^2 = 5 + 2\sqrt{6}$$

We then will show that this claim is false by remarking that the left hand side is rational, and thus we will prove the RHS is irrational, proceeding again by contradiction.

Suppose for sake of contradiction $\sqrt{6}$ is rational and can be expressed in lowest terms as the fraction $\frac{p}{q}$. Squaring both sides and rearranging yields

$$p^2 = 6q^2$$

Notice from the RHS that this means that $3 \mid p$, so $p = 3p'$ for some $p' \in \mathbb{Z}+$. Rewriting:

$$(3p')^2 = 6q^2 \to 3(p')^2 = 2q^2$$

Notice from the LHS now that $3 \mid q$, and thus we have a contradiction as 3 divides both p and q, but

$$\gcd(p, q) = 1$$

so we have a contradiction $\square$

Problem 4

Solution notes

We wish to show $|a|+|b|=|a+b| \iff ab \geq 0$

In order to do so, Let's consider the triangle inequality $|a+b| \leq |a|+|b|$ Squaring both sides, we have

$$|a+b|^2 \leq |a|^2+|b|^2+2|a||b| \iff a^2+b^2+2ab \leq a^2+b^2+2|a||b|$$

Further analyzing, we have $2ab \leq 2|a||b| \to ab \leq |a||b| \iff ab \leq |ab|$

Notice equality is only held when $ab=|ab|$ which is true when $ab \geq 0$, so we are done.

Problem 5

Solution notes

We proceed via induction, starting with the base case $n=2$. Notice by triangle inequality $|a+c| \leq |a|+|c|$, for $a, c \in \mathbb{R}$

Claim: (and base case n = 2) $|a-b| \geq |a|-|b|$

Proof: Assuming the triangle inequality, We can rewrite $|a|=|(a-b)+b| \leq |a-b|+|b| \iff |a|-|b| \leq |a-b|$

So we have proven the base case from the triangle inequality

Let's denote $P(k)$ the truth value of the inequality

$$|a_1-a_2-\cdots-a_k| \geq |a_1|-|a_2|-\cdots-|a_k|$$

Our inductive hypothesis is that for an arbitrary $k \in \mathbb{Z}^+$, $P(1), P(2), \cdots, P(k)$ are true. Notice that we wish to show that

$$|a_1-a_2-\cdots-a_{k+1}| \geq |a_1|-|a_2|-\cdots-|a_{k+1}|$$

Notice first by the base case ($|a-b| \geq |a|-|b|$, where here $a=a_1-a_2-\cdots-a_k$),

$$|a_1-a_2-\cdots a_{k+1}| \geq |a_1-a_2-\cdots-a_k|-|a_{k+1}|$$

We now plug in our inductive hypothesis, in order to write

$$|a_1 - a_2 - \cdots - a_k| \geq |a_1| - |a_2| - \cdots - |a_k|$$

Plugging this in above, we have

$$|a_1 - a_2 - \cdots - a_{k+1}| \geq |a_1 - a_2 - \cdots - a_k| - |a_{k+1}| \geq |a_1| - |a_2| - \cdots - |a_{k+1}|$$

and we are done, as we've shown $P(k) \implies P(k+1)$